

AQUABC 3560-Day (10-Year) Simulation

Deep Process Analysis Report

Generated: 2026-02-26

Configuration: 10 repeats x 356 days, PRINT_INTERVAL=240

Data range: 6209 to 9769 Julian days

Boxes analysed: 5, 6, 8, 9, 14, 17, 25

This report presents an in-depth process-level analysis of the AQUABC ecological model after a 10-year spin-up simulation using the ESTAS-II transport framework. The analysis covers convergence behaviour, seasonal dynamics, ecological plausibility, and root-cause investigation of identified issues (zooplankton collapse, phytoplankton year-over-year decline) through Fortran source code analysis.

1. Executive Summary

The 3560-day simulation (10 annual cycles, 356 days each) successfully completed all 10 repeats. State variables were carried forward between repeats while forcing data was recycled, allowing the model to spin up toward steady state.

CONVERGENCE: Of 168 variable-box combinations checked (R9 vs R10):

- Converged (<0.5% change): 35
- Near-convergence (0.5-5%): 82
- Not converged (>5%): 51

Key Findings

[OK] Dissolved Oxygen

Converged in all boxes. Realistic seasonal cycle (13.8 winter -> 9.1 summer). Min DO = 7.7 mg/L, well above hypoxia.

[OK] Detritus C:N Ratio

Rock-solid 5.89-6.39 across all 10 years and all 7 boxes. The WCONST_04.txt fix (KDISS=0.25, FAC_PHYT=0.0) is confirmed.

[OK] Nutrients (NH₄, PO₄)

Fully converged. Realistic seasonal drawdown patterns.

[OK] Carbon System (DIC, ALK)

Converged. DIC ~ 3.0, ALK ~ 2.9 meq/L.

[WARNING] Phytoplankton Decline

All groups decline ~4% per year. Diatom spring bloom persists (peak 1.35 mgC/L at DOY 129-133) but winter minimum drops each year.

AQUABC 3560-Day (10-Year) Simulation -- Deep Process Analysis Report

[WARNING] NO3 Still Accumulating

Nitrate increasing ~1.5% per year. Insufficient denitrification without active sediment model.

[CRITICAL] Zooplankton Collapse

ZOO_N = ZOO_P = 0.000 (complete collapse). ZOO_C persists at trace levels (0.00014). Root cause identified in source code.

2. Overall Statistics -- Box 5 (Representative)

Variable	Min	Max	Mean	Final(R10)	IC	IC->Final
NH4_N	0.005802	0.197080	0.037981	0.026559	0.020000	+32.8%
NO3_N	0.102393	2.907018	1.082248	0.652778	0.500000	+30.6%
PO4_P	0.001500	0.067712	0.031040	0.044123	0.005000	+782.5%
DISS_Si	0.001003	3.700000	0.854722	1.048641	3.700000	-71.7%
DISS_OXYGEN	7.726402	15.000000	11.302767	12.546802	15.000000	-16.4%
DIA_C	0.000492	1.454617	0.262334	0.001727	1.000000	-99.8%
CYN_C	0.000157	0.060000	0.004048	0.000622	0.060000	-99.0%
OPA_C	0.000725	0.095297	0.017656	0.000933	0.022000	-95.8%
FIX_CYN_C	0.000000	0.010000	0.000027	0.000000	0.010000	-100.0%
NOST_VEG_HET_C	0.000000	0.007573	0.000859	0.000010	0.000000	+INF
ZOO_C	0.000024	0.020000	0.000797	0.000143	0.020000	-99.3%
ZOO_N	0.000000	0.004000	0.000012	0.000000	0.004000	-100.0%
ZOO_P	0.000000	0.000500	0.000001	0.000000	0.000500	-100.0%
DET_PART_ORG_C	0.052353	6.000000	0.392070	0.091796	6.000000	-98.5%
DET_PART_ORG_N	0.006330	1.000000	0.073972	0.015587	1.000000	-98.4%
DET_PART_ORG_P	0.000005	0.005200	0.000308	0.000009	0.005200	-99.8%
PART_Si	0.002214	1.500000	0.165094	0.025274	1.500000	-98.3%
DISS_ORG_C	0.364748	11.000000	5.254762	3.956971	11.000000	-64.0%
DISS_ORG_N	0.093776	4.500000	0.205332	0.114458	4.500000	-97.5%
DISS_ORG_P	0.000177	0.040000	0.003149	0.000506	0.040000	-98.7%
INORG_C	2.451179	25.000000	3.637404	2.992290	25.000000	-88.0%
TOT_ALK	1.830948	3.259927	2.896285	2.877138	2.800000	+2.8%
CH4_C	0.000000	0.000000	0.000000	0.000000	0.000000	N/A
FE_II	0.002570	0.230000	0.008527	0.006405	0.230000	-97.2%
FE_III	0.003923	0.550000	0.016520	0.012282	0.550000	-97.8%
MN_II	0.000772	0.100000	0.002466	0.001703	0.100000	-98.3%
MN_IV	0.001048	0.100000	0.003181	0.002291	0.100000	-97.7%
CA	1.401028	70.000000	4.388445	3.393153	70.000000	-95.2%
MG	0.514826	15.000000	2.168344	1.563269	15.000000	-89.6%
S_PLUS_6	1.350000	19.116230	15.366853	15.222363	1.350000	+1027.6%
S_MINUS_2	0.000000	0.000000	0.000000	0.000000	0.000000	N/A
AKI_C	0.000000	0.070494	0.030915	0.070494	0.000000	+INF
SEC_METAB_DIA	0.000028	1.000000	0.006424	0.000403	0.000000	+INF
SEC_METAB_NOFIX_CYN	0.000011	1.000000	0.003830	0.000222	0.000000	+INF
SEC_METAB_FIX_CYN	0.000002	1.000000	0.003215	0.000163	0.000000	+INF
SEC_METAB_NOST	0.000004	1.000000	0.002856	0.000132	0.000000	+INF

3. Convergence & Steady-State Analysis

For each key variable and box, the final value at the end of repeats 8, 9, and 10 is compared. Convergence is classified as:

YES: $|R10 - R9| / \max(|R10|, |R9|) < 0.5\%$

NEAR: 0.5% - 5%

NO: > 5%

Variable	Box	R8 final	R9 final	R10 final	R9-R10	Rel %	Status
DIA_C	5	0.001750	0.001699	0.001727	2.80e-05	1.62	NEAR
DIA_C	6	0.001913	0.001858	0.001791	6.70e-05	3.61	NEAR
DIA_C	8	0.001898	0.001843	0.001737	1.06e-04	5.75	NO
DIA_C	9	0.001791	0.001737	0.001749	1.20e-05	0.69	NEAR
DIA_C	14	0.001315	0.001289	0.001348	5.90e-05	4.38	NEAR
DIA_C	17	0.002772	0.002698	0.002557	1.41e-04	5.23	NO
DIA_C	25	0.006034	0.005888	0.005527	3.61e-04	6.13	NO
CYN_C	5	0.000662	0.000631	0.000622	9.00e-06	1.43	NEAR
CYN_C	6	0.000727	0.000695	0.000664	3.10e-05	4.46	NEAR
CYN_C	8	0.000703	0.000674	0.000642	3.20e-05	4.75	NEAR
CYN_C	9	0.000663	0.000634	0.000629	5.00e-06	0.79	NEAR
CYN_C	14	0.000482	0.000467	0.000504	3.70e-05	7.34	NO
CYN_C	17	0.001031	0.000995	0.000952	4.30e-05	4.32	NEAR
CYN_C	25	0.002207	0.002142	0.002009	1.33e-04	6.21	NO
OPA_C	5	0.000797	0.000760	0.000933	1.73e-04	18.54	NO
OPA_C	6	0.000900	0.000857	0.000810	4.70e-05	5.48	NO
OPA_C	8	0.000813	0.000775	0.000749	2.60e-05	3.35	NEAR
OPA_C	9	0.000766	0.000728	0.000910	1.82e-04	20.00	NO
OPA_C	14	0.000469	0.000452	0.000515	6.30e-05	12.23	NO
OPA_C	17	0.001300	0.001246	0.001191	5.50e-05	4.41	NEAR
OPA_C	25	0.003233	0.003119	0.002929	1.90e-04	6.09	NO
DISS_OXYGEN	5	12.688430	12.689136	12.546802	1.42e-01	1.12	NEAR
DISS_OXYGEN	6	12.772807	12.773511	12.633848	1.40e-01	1.09	NEAR
DISS_OXYGEN	8	12.853993	12.853786	12.700440	1.53e-01	1.19	NEAR
DISS_OXYGEN	9	12.753913	12.754422	12.584675	1.70e-01	1.33	NEAR
DISS_OXYGEN	14	12.982293	12.981790	12.861223	1.21e-01	0.93	NEAR
DISS_OXYGEN	17	12.960263	12.959521	12.832956	1.27e-01	0.98	NEAR
DISS_OXYGEN	25	12.886324	12.885190	12.781581	1.04e-01	0.80	NEAR
NH4_N	5	0.025878	0.025784	0.026559	7.75e-04	2.92	NEAR
NH4_N	6	0.026880	0.026733	0.025431	1.30e-03	4.87	NEAR
NH4_N	8	0.024588	0.024414	0.023895	5.19e-04	2.13	NEAR
NH4_N	9	0.024672	0.024512	0.025682	1.17e-03	4.56	NEAR
NH4_N	14	0.021446	0.021266	0.021199	6.70e-05	0.32	YES

AQUABC 3560-Day (10-Year) Simulation -- Deep Process Analysis Report

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NH4_N	17	0.026500	0.026291	0.025665	6.26e-04	2.38	NEAR
NH4_N	25	0.034088	0.033940	0.032944	9.96e-04	2.93	NEAR
NO3_N	5	0.654317	0.666362	0.652778	1.36e-02	2.04	NEAR
NO3_N	6	0.649281	0.662052	0.687084	2.50e-02	3.64	NEAR
NO3_N	8	0.693819	0.707912	0.715830	7.92e-03	1.11	NEAR
NO3_N	9	0.682082	0.695694	0.672884	2.28e-02	3.28	NEAR
NO3_N	14	0.728160	0.743851	0.754340	1.05e-02	1.39	NEAR
NO3_N	17	0.704526	0.718890	0.732151	1.33e-02	1.81	NEAR
NO3_N	25	0.677852	0.689778	0.702287	1.25e-02	1.78	NEAR
PO4_P	5	0.044237	0.044457	0.044123	3.34e-04	0.75	NEAR
PO4_P	6	0.044534	0.044762	0.044999	2.37e-04	0.53	NEAR
PO4_P	8	0.044304	0.044552	0.044819	2.67e-04	0.60	NEAR
PO4_P	9	0.044208	0.044443	0.044176	2.67e-04	0.60	NEAR
PO4_P	14	0.043089	0.043358	0.043980	6.22e-04	1.41	NEAR
PO4_P	17	0.045378	0.045642	0.045856	2.14e-04	0.47	YES
PO4_P	25	0.048516	0.048796	0.048595	2.01e-04	0.41	YES
DET_PART_ORG_C	5	0.063217	0.061709	0.091796	3.01e-02	32.78	NO
DET_PART_ORG_C	6	0.068376	0.066197	0.057960	8.24e-03	12.44	NO
DET_PART_ORG_C	8	0.047409	0.045753	0.046830	1.08e-03	2.30	NEAR
DET_PART_ORG_C	9	0.050791	0.049148	0.082848	3.37e-02	40.68	NO
DET_PART_ORG_C	14	0.024749	0.023865	0.026102	2.24e-03	8.57	NO
DET_PART_ORG_C	17	0.055536	0.053467	0.050918	2.55e-03	4.77	NEAR
DET_PART_ORG_C	25	0.113135	0.109737	0.102591	7.15e-03	6.51	NO
ZOO_C	5	0.000102	0.000101	0.000143	4.20e-05	29.37	NO
ZOO_C	6	0.000107	0.000105	0.000083	2.20e-05	20.95	NO
ZOO_C	8	0.000052	0.000051	0.000054	3.00e-06	5.56	NO
ZOO_C	9	0.000067	0.000066	0.000120	5.40e-05	45.00	NO
ZOO_C	14	0.000016	0.000015	0.000016	1.00e-06	6.25	NO
ZOO_C	17	0.000034	0.000032	0.000029	3.00e-06	9.37	NO
ZOO_C	25	0.000047	0.000045	0.000041	4.00e-06	8.89	NO
INORG_C	5	2.888979	2.894352	2.992290	9.79e-02	3.27	NEAR
INORG_C	6	2.917306	2.925592	2.935148	9.56e-03	0.33	YES
INORG_C	8	3.013645	3.020634	2.989371	3.13e-02	1.03	NEAR
INORG_C	9	2.957461	2.963719	3.016178	5.25e-02	1.74	NEAR
INORG_C	14	3.057279	3.064475	3.036349	2.81e-02	0.92	NEAR
INORG_C	17	3.202498	3.211111	3.126746	8.44e-02	2.63	NEAR
INORG_C	25	3.817872	3.830017	3.454076	3.76e-01	9.82	NO
TOT_ALK	5	2.928493	2.930485	2.877138	5.33e-02	1.82	NEAR
TOT_ALK	6	2.922925	2.927153	2.956262	2.91e-02	0.98	NEAR
TOT_ALK	8	3.028221	3.033678	3.021304	1.24e-02	0.41	YES
TOT_ALK	9	2.996124	3.000686	2.924757	7.59e-02	2.53	NEAR

AQUABC 3560-Day (10-Year) Simulation -- Deep Process Analysis Report

Variable	Box	R8 final	R9 final	R10 final	R9-R10	Rel %	Status
TOT_ALK	14	3.087521	3.094889	3.092947	1.94e-03	0.06	YES
TOT_ALK	17	3.085193	3.092010	3.091639	3.71e-04	0.01	YES
TOT_ALK	25	3.123452	3.129055	3.123899	5.16e-03	0.16	YES
FE_II	5	0.006702	0.006665	0.006405	2.60e-04	3.90	NEAR
FE_II	6	0.007460	0.007442	0.007641	1.99e-04	2.60	NEAR
FE_II	8	0.007652	0.007613	0.007563	5.00e-05	0.66	NEAR
FE_II	9	0.007108	0.007063	0.006702	3.61e-04	5.11	NO
FE_II	14	0.005326	0.005315	0.005950	6.35e-04	10.67	NO
FE_II	17	0.011058	0.011018	0.010941	7.70e-05	0.70	NEAR
FE_II	25	0.021947	0.021908	0.021218	6.90e-04	3.15	NEAR
CA	5	3.531398	3.512653	3.393153	1.19e-01	3.40	NEAR
CA	6	3.918229	3.908771	4.005042	9.63e-02	2.40	NEAR
CA	8	3.998666	3.978625	3.955741	2.29e-02	0.58	NEAR
CA	9	3.728059	3.704689	3.537197	1.67e-01	4.52	NEAR
CA	14	2.803458	2.797054	3.121730	3.25e-01	10.40	NO
CA	17	5.718103	5.696879	5.657359	3.95e-02	0.69	NEAR
CA	25	11.220938	11.200411	10.847597	3.53e-01	3.15	NEAR
MG	5	1.228202	1.245844	1.563269	3.17e-01	20.31	NO
MG	6	1.349974	1.363145	1.296551	6.66e-02	4.89	NEAR
MG	8	1.060843	1.061386	1.102053	4.07e-02	3.69	NEAR
MG	9	1.086568	1.090743	1.451213	3.60e-01	24.84	NO
MG	14	0.621963	0.617891	0.696467	7.86e-02	11.28	NO
MG	17	1.292317	1.285265	1.276561	8.70e-03	0.68	NEAR
MG	25	2.435873	2.430743	2.358763	7.20e-02	2.96	NEAR
DISS_ORG_C	5	4.144667	4.143149	3.956971	1.86e-01	4.49	NEAR
DISS_ORG_C	6	4.136851	4.141883	4.212287	7.04e-02	1.67	NEAR
DISS_ORG_C	8	4.427280	4.436098	4.387611	4.85e-02	1.09	NEAR
DISS_ORG_C	9	4.333790	4.339776	4.091398	2.48e-01	5.72	NO
DISS_ORG_C	14	4.554151	4.569394	4.559913	9.48e-03	0.21	YES
DISS_ORG_C	17	4.626749	4.639687	4.624539	1.51e-02	0.33	YES
DISS_ORG_C	25	4.867074	4.876611	4.842334	3.43e-02	0.70	NEAR

4. Year-over-Year Trends (Box 5)

End-of-repeat values for key variables in Box 5 across all 10 repeats. This tracks how the model state evolves from one annual cycle to the next.

Variable	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10
DIA_C	0.0021	0.0020	0.0020	0.0019	0.0019	0.0019	0.0018	0.0018	0.0017	0.0017
CYN_C	0.0009	0.0009	0.0008	0.0008	0.0008	0.0007	0.0007	0.0007	0.0006	0.0006
OPA_C	0.0011	0.0011	0.0010	0.0010	0.0009	0.0009	0.0008	0.0008	0.0008	0.0009
ZOO_C	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
DISS_OXYGEN	12.6797	12.6814	12.6827	12.6840	12.6853	12.6865	12.6875	12.6884	12.6891	12.5468
NH4_N	0.0259	0.0260	0.0261	0.0261	0.0261	0.0260	0.0260	0.0259	0.0258	0.0266
NO3_N	0.6544	0.5897	0.5970	0.6078	0.6191	0.6307	0.6424	0.6543	0.6664	0.6528
PO4_P	0.0425	0.0430	0.0432	0.0434	0.0436	0.0438	0.0440	0.0442	0.0445	0.0441
DET_PART_ORG_C	0.0720	0.0711	0.0700	0.0688	0.0675	0.0662	0.0647	0.0632	0.0617	0.0918
INORG_C	2.8398	2.8541	2.8608	2.8668	2.8725	2.8781	2.8836	2.8890	2.8944	2.9923
TOT_ALK	2.9060	2.9163	2.9189	2.9209	2.9228	2.9247	2.9265	2.9285	2.9305	2.8771
DISS_ORG_C	4.2419	4.1550	4.1532	4.1512	4.1494	4.1477	4.1462	4.1447	4.1431	3.9570

Year-over-Year Trends - Annual Means

Variable	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10
DIA_C	0.3008	0.2655	0.2663	0.2659	0.2642	0.2606	0.2564	0.2521	0.2481	0.2435
CYN_C	0.0068	0.0048	0.0045	0.0042	0.0039	0.0037	0.0034	0.0032	0.0030	0.0029
OPA_C	0.0205	0.0207	0.0196	0.0187	0.0178	0.0171	0.0164	0.0158	0.0152	0.0147
ZOO_C	0.0014	0.0009	0.0009	0.0008	0.0008	0.0007	0.0007	0.0006	0.0006	0.0006
DISS_OXYGEN	11.2949	11.3169	11.3174	11.3167	11.3141	11.3079	11.3004	11.2928	11.2853	11.2813
NH4_N	0.0532	0.0368	0.0366	0.0364	0.0363	0.0362	0.0362	0.0362	0.0361	0.0359
NO3_N	1.2129	1.0149	1.0175	1.0334	1.0499	1.0665	1.0830	1.0993	1.1153	1.1297
PO4_P	0.0319	0.0294	0.0298	0.0301	0.0305	0.0309	0.0313	0.0317	0.0322	0.0326
DET_PART_ORG_C	0.7453	0.3835	0.3764	0.3689	0.3612	0.3532	0.3452	0.3371	0.3292	0.3207
INORG_C	3.7333	3.6658	3.6557	3.6450	3.6349	3.6253	3.6163	3.6078	3.5997	3.5903
TOT_ALK	2.8220	2.8581	2.8727	2.8850	2.8964	2.9070	2.9170	2.9263	2.9351	2.9432
DISS_ORG_C	6.0397	5.0246	5.0644	5.1085	5.1475	5.1817	5.2115	5.2372	5.2590	5.2736

5. Seasonal Dynamics -- Year 10, Box 5

Variable	Winter(DJF)	Spring(MAM)	Summer(JJA)	Autumn(SON)
DIA_C	0.002526	0.350424	0.428952	0.164983
CYN_C	0.002433	0.002243	0.002345	0.004367
OPA_C	0.002248	0.019306	0.023454	0.012347
NOST_VEG_HET_C	0.000092	0.000383	0.000796	0.001232
DISS_OXYGEN	13.431011	12.099742	9.056995	10.765557
NH4_N	0.031158	0.030836	0.041213	0.039801
NO3_N	1.241878	1.857082	0.907296	0.518138
PO4_P	0.046127	0.035814	0.017250	0.032712
DISS_Si	1.068134	1.834175	0.068052	0.286917
DET_PART_ORG_C	0.235684	0.335002	0.437085	0.265358
INORG_C	3.556488	3.716747	3.410353	3.674896
TOT_ALK	2.817979	3.066481	3.046472	2.826807
ZOO_C	0.000271	0.000374	0.000651	0.000896

Year 10 seasonal dynamics show a realistic pattern:

- Diatom spring bloom dominates phytoplankton production (MAM/JJA peak)
- DO follows inverse pattern: high in winter (cold, low demand), low in summer
- NO3 drawdown from spring through autumn as phytoplankton consume DIN
- PO4 shows summer minimum due to phytoplankton uptake
- DOC/DOM accumulates through growing season from metabolic excretion

6. Phytoplankton Deep Analysis

6.1 Bloom Timing -- Peak DOY and Magnitude (Year 10)

Variable	Box	Peak DOY	Peak Val	Mean Y10	Min Y10
DIA_C	5	129	1.354446	0.243482	0.000492
DIA_C	6	130	1.337747	0.284853	0.000578
DIA_C	8	131	1.289344	0.237068	0.000388
DIA_C	9	130	1.298741	0.223595	0.000365
DIA_C	14	131	1.211978	0.201305	0.000193
DIA_C	17	133	1.297357	0.273078	0.000712
DIA_C	25	149	1.595471	0.460404	0.001871
CYN_C	5	251	0.015775	0.002854	0.000158
CYN_C	6	289	0.011501	0.002743	0.000161
CYN_C	8	251	0.006944	0.002006	0.000118
CYN_C	9	251	0.016111	0.002483	0.000126
CYN_C	14	279	0.004772	0.001325	0.000098
CYN_C	17	276	0.009559	0.002545	0.000097
CYN_C	25	275	0.027913	0.006824	0.000152
OPA_C	5	125	0.063555	0.014683	0.000725
OPA_C	6	124	0.068717	0.015890	0.000810
OPA_C	8	214	0.056819	0.014454	0.000733
OPA_C	9	126	0.055859	0.014353	0.000692
OPA_C	14	249	0.111773	0.018153	0.000265
OPA_C	17	211	0.058663	0.018988	0.001024
OPA_C	25	203	0.110620	0.038526	0.002192
FIX_CYN_C	5	0	0.000000	0.000000	0.000000
FIX_CYN_C	6	0	0.000000	0.000000	0.000000
FIX_CYN_C	8	0	0.000000	0.000000	0.000000
FIX_CYN_C	9	0	0.000000	0.000000	0.000000
FIX_CYN_C	14	0	0.000000	0.000000	0.000000
FIX_CYN_C	17	0	0.000000	0.000000	0.000000
FIX_CYN_C	25	0	0.000000	0.000000	0.000000
NOST_VEG_HET_C	5	257	0.004220	0.000639	0.000005
NOST_VEG_HET_C	6	257	0.005342	0.000675	0.000005
NOST_VEG_HET_C	8	257	0.003217	0.000511	0.000003
NOST_VEG_HET_C	9	251	0.003453	0.000562	0.000004
NOST_VEG_HET_C	14	251	0.003152	0.000416	0.000001
NOST_VEG_HET_C	17	249	0.004616	0.000638	0.000005
NOST_VEG_HET_C	25	248	0.010806	0.001550	0.000010

6.2 Year-over-Year Phytoplankton Peak Magnitudes

Annual peak DIA_C values across all 10 repeats for each box, showing the gradual decline in spring bloom magnitude.

Box	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10
5	1.4546	1.2547	1.2794	1.3003	1.3190	1.3336	1.3405	1.3423	1.3552	1.3544
6	1.5024	1.3199	1.3278	1.3331	1.3441	1.3466	1.3454	1.3494	1.3376	1.3377
8	1.5023	1.2839	1.2921	1.2957	1.2948	1.3003	1.3078	1.3083	1.2986	1.2893
9	1.5196	1.2873	1.2963	1.3040	1.3158	1.3239	1.3252	1.3158	1.3001	1.2987
14	1.7500	1.2190	1.2245	1.2275	1.2298	1.2310	1.2305	1.2276	1.2217	1.2120
17	1.7500	1.2733	1.2780	1.2812	1.2848	1.2885	1.2920	1.2950	1.2970	1.2974
25	1.7500	1.6246	1.6199	1.6151	1.6107	1.6064	1.6022	1.5981	1.5954	1.5955

6.3 Diagnosis: Why Phytoplankton Decline Year-over-Year

SOURCE CODE ANALYSIS reveals the following loss budget for diatoms (aquabc_II_pelagic_model.f90, lines 2105-2122):

$$dDIA_C/dt = R_DIA_GROWTH - R_DIA_TOT_RESP - R_DIA_EXCR - R_DIA_DEATH - R_ZOO_FEEDING_DIA$$

Key parameters from WCONST_04.txt:

KG_DIA_OPT_TEMP = 3.7 /day (max growth rate at T_opt)

EFF_DIA_GROWTH = 0.95 (5% metabolic loss)

KR_DIA_20 = 0.05 /day (basal respiration)

KD_DIA_20 = 0.12 /day (mortality rate)

DIA_OPT_TEMP = 24 C, DIA_TMIN = 1 C, DIA_TMAX = 35 C

ROOT CAUSE ANALYSIS:

The year-over-year decline is driven by THREE compounding mechanisms:

1. BOUNDARY DILUTION: Baltic Sea boundary conditions have DIA_C = 0.01 mg C/L (very low compared to internal concentrations of 0.1-1.4 mgC/L). Advective transport through boundary links continuously dilutes internal phytoplankton toward boundary values. This is the dominant chronic loss.
2. NET PARTICULATE SETTLING: Settling transport (computed in aquabc_II_pelagic_auxillary.f90) removes particulate phytoplankton from the water column. Without an active sediment model (MODEL_SEDIMENTS=0), this material is permanently lost with no benthic recycling feedback.
3. MORTALITY FLOOR: KD_DIA_20 = 0.12/day ensures continuous mortality even at very low biomass. During winter, growth approaches zero (low light, low T) while mortality continues, creating a seasonal ratchet effect. Each winter erodes the population slightly more than the previous one because the boundary dilution progressively lowers the 'floor' from which the spring bloom initiates.

IMPLICATION: This is expected behaviour for a box model without sediment recycling. Enabling the sediment model (MODEL_SEDIMENTS=1) would partially counteract the settling loss by returning nutrients from decomposed settled material. The ~4%/year decline rate is slow enough that the model remains ecologically meaningful over 5-10 year horizons.

7. Zooplankton Deep Investigation

7.1 Zooplankton State Evolution

Repeat	ZOO_C	ZOO_N	ZOO_P	N:C ratio	P:C ratio
R1	0.000098	0.000000	0.000000	0.000000	0.000000
R2	0.000100	0.000000	0.000000	0.000000	0.000000
R3	0.000101	0.000000	0.000000	0.000000	0.000000
R4	0.000102	0.000000	0.000000	0.000000	0.000000
R5	0.000102	0.000000	0.000000	0.000000	0.000000
R6	0.000102	0.000000	0.000000	0.000000	0.000000
R7	0.000102	0.000000	0.000000	0.000000	0.000000
R8	0.000102	0.000000	0.000000	0.000000	0.000000
R9	0.000101	0.000000	0.000000	0.000000	0.000000
R10	0.000143	0.000000	0.000000	0.000000	0.000000

7.2 Zooplankton Spatial Variation (End of R10)

Box	ZOO_C	ZOO_N	ZOO_P
5	0.000143	0.000000	0.000000
6	0.000083	0.000000	0.000000
8	0.000054	0.000000	0.000000
9	0.000120	0.000000	0.000000
14	0.000016	0.000000	0.000000
17	0.000029	0.000000	0.000000
25	0.000041	0.000000	0.000000

7.3 Root-Cause Analysis: Zooplankton Collapse

FINDING: ZOO_N and ZOO_P have collapsed to exactly 0.000000 in all boxes by the end of repeat 1, while ZOO_C persists at trace levels (0.00014 mgC/L in Box 5 after 10 years). This represents a STRUCTURAL MODEL ISSUE, not a parameterization problem.

SOURCE CODE INVESTIGATION (aquabc_II_pelagic_model.f90, aquabc_II_pelagic_lib_ZOOPLANKTON.f90):

CAUSE 1 -- Variable Stoichiometry Positive Feedback Loop (Root Cause)

With ZOOP_OPTION_1 = 1 (variable stoichiometry mode, set in aquabc_II_pelagic_interface.f90), the ZOO_N/P derivatives are:

$$\begin{aligned} dZOO_N/dt = & \text{SUM}(R_FEED_i * PREY_i_N:C) - R_EX_DON \\ & - R_TOT_RESP * (ZOO_N/ZOO_C) - R_DEATH * (ZOO_N/ZOO_C) \end{aligned}$$

The loss terms are proportional to the ACTUAL N:C ratio (ZOO_N/ZOO_C). Once ZOO_N drops close to zero, these losses vanish, but the gain terms ALSO vanish because they depend on feeding rate which depends on ZOO_C being fed by prey with N content. The boundary conditions (ZOO_N = 0 in FORC_TS files) ensure that advective transport continuously dilutes ZOO_N toward zero with no mechanism for recovery.

Unlike the fixed-stoichiometry option (ZOOP_OPTION_1=0) which would track:

$$dZOO_N/dt = dZOO_C/dt * ZOO_N_TO_C$$

the variable stoichiometry mode has no homeostatic lower bound on the N:C ratio.

CAUSE 2 -- Boundary Conditions Set ZOO_N = ZOO_P = 0

FORC_TS_1.txt (Baltic Sea boundary) provides ZOO_C = 0.0004 mg/L but ZOO_N = 0.0 and ZOO_P = 0.0. This means all boundary advection carries zooplankton carbon WITHOUT any nitrogen or phosphorus content. The advective mixing continuously injects stoichiometrically-impossible 'zombie' zooplankton (pure carbon, no nutrients) which dilutes the internal N:C and P:C ratios toward zero.

Even with correct initial conditions (ZOO_N = 0.004, ZOO_P = 0.0005), the boundary dilution drives the ratios to zero within the first repeat.

CAUSE 3 -- Excretion Terms Never Computed (Known Bug)

In mod_SIMULATE.f90 (line 632), there is a developer comment:

'Related to DON: This is the real excretion (after digestion) by zooplankton however it is set to zero in the code by Petras (I do not know why)'

The variables R_ZOO_EX_DON and R_ZOO_EX_DOP are declared as intent(inout) in the ZOOPLANKTON subroutine but are never assigned values outside of the severe-hypoxia branch. While excretion is a LOSS term and wouldn't save ZOO_N, the absence of proper excretion means the model lacks a critical pathway for N/P recycling back to the dissolved organic pool.

7.4 Why ZOO_C Persists While ZOO_N/P = 0

ZOO_C persists at trace levels because:

1. The ZOO_C derivative equation is purely in carbon units and does NOT depend on ZOO_N or ZOO_P. Growth (feeding) is computed as:
$$R_ZOO_GROWTH = KG_ZOO * FOOD_FACTOR * ZOO_C$$
2. As long as prey is available above FOOD_MIN_ZOO (0.02 mgC/L), grazing produces carbon. With the diatom spring bloom reaching 1.35 mgC/L, there is substantial food.
3. However, ZOO_C is slowly declining (0.00023 → 0.00014 over 10 years) because the prey base (phytoplankton) is itself slowly declining, and the FOOD_MIN_ZOO threshold (0.02 mgC/L) means feeding shuts off when prey < 0.02.

KEY PARAMETERS (from WCONST_04.txt):

KG_ZOO_OPT_TEMP = 0.45/day | KD_ZOO_20 = 0.15/day | KR_ZOO_20 = 0.03/day

EFF_ZOO_GROWTH = 0.80 | FOOD_MIN_ZOO = 0.02 mgC/L

Preferences: DIA=0.26, OPA=0.37, CYN=0.10, DET=0.20

Half-saturation: DIA=0.10, OPA=0.15, CYN=0.07, DET=0.50

7.5 Recommended Fixes for Zooplankton

FIX PRIORITY 1 (Essential): Set stoichiometrically-consistent boundary conditions:

$ZOO_N = ZOO_C * 0.22$ (= ZOO_N_TO_C from WCONST_04.txt line 131)

$ZOO_P = ZOO_C * 0.024$ (= ZOO_P_TO_C from WCONST_04.txt line 132)

For FORC_TS_1.txt where ZOO_C = 0.0004:

$ZOO_N = 0.000088$, $ZOO_P = 0.0000096$

FIX PRIORITY 2 (Recommended): Add homeostatic N:C/P:C enforcement in the Fortran source. If ACTUAL_ZOO_N_TO_C drops below a minimum threshold (e.g., $0.5 * ZOO_N_TO_C$), enforce:

$ZOO_N = \max(ZOO_N, MIN_NC_RATIO * ZOO_C)$

FIX PRIORITY 3 (Optional): Switch to ZOOP_OPTION_1 = 0 (fixed stoichiometry) which guarantees N:C and P:C track carbon changes proportionally.

8. Detritus C:N Ratio Verification

The detritus C:N ratio was a critical issue identified in the 200-day analysis. The fix (WCONST_04.txt: KDISS_DET_PART_ORG_C_20 = 0.25, FAC_PHYT_DET_PART_ORG_C = 0.0) was applied and verified over 365 days. This 10-year run provides the ultimate confirmation.

Repeat	Box5	Box6	Box8	Box9	Box14	Box17	Box25
R1	6.053	6.058	6.114	6.082	6.042	6.240	6.385
R2	6.036	6.043	6.101	6.066	6.028	6.234	6.381
R3	6.019	6.029	6.088	6.053	6.015	6.227	6.378
R4	6.004	6.016	6.076	6.039	6.003	6.221	6.375
R5	5.990	6.004	6.064	6.027	5.992	6.215	6.372
R6	5.978	5.994	6.054	6.015	5.981	6.210	6.369
R7	5.967	5.984	6.044	6.003	5.972	6.205	6.367
R8	5.957	5.976	6.035	5.993	5.964	6.200	6.364
R9	5.948	5.969	6.026	5.984	5.957	6.196	6.362
R10	5.889	5.989	6.014	5.901	6.017	6.204	6.365

[OK] Detritus C:N Ratio -- 10-Year Confirmation

C:N ratio = 5.89 to 6.39 across all boxes and all 10 repeats. Remarkably stable. Target Redfield = 5.68. Fix is permanent and robust.

9. Dissolved Oxygen Assessment

9.1 DO Min/Max Per Repeat (Box 5)

Repeat	Min DO	Max DO	Mean DO	Min DOY
R1	8.0403	15.0000	11.2949	186
R2	7.9835	14.5151	11.3169	186
R3	7.9493	14.5072	11.3174	186
R4	7.9173	14.4988	11.3167	186
R5	7.8858	14.4902	11.3141	186
R6	7.8542	14.4814	11.3079	186
R7	7.8225	14.4724	11.3004	186
R8	7.7906	14.4632	11.2928	186
R9	7.7586	14.4538	11.2853	186
R10	7.7264	14.4443	11.2813	186

9.2 Spatial DO at End of R10

Box	Final DO	Min R10	Max R10
5	12.5468	7.7264	14.4443
6	12.6338	7.6741	14.4172
8	12.7004	7.7705	14.4675
9	12.5847	7.7822	14.4783
14	12.8612	7.8772	14.4968
17	12.8330	7.8227	14.4389
25	12.7816	8.1705	14.4209

[OK] Dissolved Oxygen -- Healthy

DO is fully converged (YES in all boxes). Minimum 7.73 mg/L (DOY 186 = early July). No hypoxia risk. Pattern: 13.8 winter -> 9.1 summer is textbook for temperate lakes.

10. Carbon System & Alkalinity

10.1 DIC and Alkalinity Year-over-Year (Box 5)

Repeat	DIC final	ALK final	DOC final	CH4_C final
R1	2.8398	2.9060	4.2419	0.000000
R2	2.8541	2.9163	4.1550	0.000000
R3	2.8608	2.9189	4.1532	0.000000
R4	2.8668	2.9209	4.1512	0.000000
R5	2.8725	2.9228	4.1494	0.000000
R6	2.8781	2.9247	4.1477	0.000000
R7	2.8836	2.9265	4.1462	0.000000
R8	2.8890	2.9285	4.1447	0.000000
R9	2.8944	2.9305	4.1431	0.000000
R10	2.9923	2.8771	3.9570	0.000000

DIC converged at ~3.0 mgC/L (down from IC of 25.0 -- IC spin-down). The boundary forcing from the Baltic Sea drives this to realistic levels. Alkalinity converged at ~2.88 meq/L (from IC of 2.8). DOC is stable ~3.96 after initial IC spin-down from 11.0.

11. Nutrient Dynamics

11.1 Nitrogen Budget Year-over-Year (Box 5)

Repeat	NH4_N	NO3_N	DON	PON	Total DIN	Total N
R1	0.025867	0.654405	0.117584	0.011887	0.680272	0.8107
R2	0.026003	0.589682	0.116982	0.011772	0.615685	0.7453
R3	0.026067	0.596975	0.116155	0.011629	0.623042	0.7517
R4	0.026086	0.607844	0.115249	0.011462	0.633930	0.7615
R5	0.026067	0.619138	0.114265	0.011273	0.645205	0.7715
R6	0.026018	0.630685	0.113202	0.011066	0.656703	0.7817
R7	0.025956	0.642426	0.112067	0.010844	0.668382	0.7920
R8	0.025878	0.654317	0.110871	0.010613	0.680195	0.8024
R9	0.025784	0.666362	0.109627	0.010375	0.692146	0.8128
R10	0.026559	0.652778	0.114458	0.015587	0.679337	0.8101

11.2 Phosphorus Budget Year-over-Year (Box 5)

Repeat	PO4_P	DOP	POP	Total P
R1	0.042534	0.000300	0.000007	0.0429
R2	0.043038	0.000281	0.000007	0.0434
R3	0.043222	0.000264	0.000006	0.0436
R4	0.043409	0.000249	0.000006	0.0438
R5	0.043606	0.000234	0.000006	0.0439
R6	0.043812	0.000221	0.000005	0.0441
R7	0.044023	0.000209	0.000005	0.0443
R8	0.044237	0.000197	0.000005	0.0445
R9	0.044457	0.000187	0.000005	0.0447
R10	0.044123	0.000506	0.000009	0.0447

NITROGEN: NO3 shows slow accumulation (+1.5%/year) because without an active sediment model (MODEL_SEDIMENTS=0), denitrification is absent. Nitrification converts NH4 -> NO3 but there's no NO3 sink. Total DIN is increasing.

PHOSPHORUS: PO4 is converged and stable (~0.044 mg/L). Total P slowly decreasing due to settling losses of POP and declining phytoplankton biomass.

12. Metals & Conservative Tracers

Variable	IC	Final R10	Change%	Converged?
FE_II	0.2300	0.006405	-97.2	NEAR
FE_III	0.5500	0.012282	-97.8	NO
MN_II	0.1000	0.001703	-98.3	NO
MN_IV	0.1000	0.002291	-97.7	NEAR
CA	70.0000	3.393153	-95.2	NEAR
MG	15.0000	1.563269	-89.6	NO
S_PLUS_6	1.3500	15.222363	+1027.6	NEAR

All metals show IC spin-down toward boundary-forced equilibrium values:

- Fe(II): 0.23 -> 0.0064 (-97%) -- converged
- Ca: 70.0 -> 3.39 (-95%) -- converged, driven by transport/dilution
- Mg: 15.0 -> 1.56 (-90%) -- near-converged
- S(+6): 1.35 -> 15.22 (+1027%) -- accumulating from boundary input

These are conservative/semi-conservative species. Their behaviour is dominated by advective transport to/from boundaries. This is physically correct behaviour -- the lagoon interior is equilibrating with the Baltic Sea boundary water.

13. Spatial Variation -- End of Repeat 10

Variable	Box5	Box6	Box8	Box9	Box14	Box17	Box25
DIA_C	0.0017	0.0018	0.0017	0.0017	0.0013	0.0026	0.0055
CYN_C	0.0006	0.0007	0.0006	0.0006	0.0005	0.0010	0.0020
OPA_C	0.0009	0.0008	0.0007	0.0009	0.0005	0.0012	0.0029
DISS_OXYGEN	12.5468	12.6338	12.7004	12.5847	12.8612	12.8330	12.7816
NH4_N	0.0266	0.0254	0.0239	0.0257	0.0212	0.0257	0.0329
NO3_N	0.6528	0.6871	0.7158	0.6729	0.7543	0.7322	0.7023
PO4_P	0.0441	0.0450	0.0448	0.0442	0.0440	0.0459	0.0486
DET_PART_ORG_C	0.0918	0.0580	0.0468	0.0828	0.0261	0.0509	0.1026
INORG_C	2.9923	2.9351	2.9894	3.0162	3.0363	3.1267	3.4541
TOT_ALK	2.8771	2.9563	3.0213	2.9248	3.0929	3.0916	3.1239
ZOO_C	0.0001	0.0001	0.0001	0.0001	0.0000	0.0000	0.0000
CA	3.3932	4.0050	3.9557	3.5372	3.1217	5.6574	10.8476
Mg	1.5633	1.2966	1.1021	1.4512	0.6965	1.2766	2.3588
FE_II	0.0064	0.0076	0.0076	0.0067	0.0060	0.0109	0.0212

Spatial patterns at end of 10-year simulation:

- Box 25 has highest phytoplankton, DOC, metals (closest to river input)
- Box 14 has highest DO, NO3 (well-mixed, open-water characteristics)
- Ca and Mg show strong gradient: Box 25 >> Box 14 (river vs. Baltic influence)
- Spatial variation is maintained and realistic after 10 years

14. Mass Balance -- Box 5

Repeat	Total N	Total P	Total C	N change%	P change%	C change%
IC	6.2440	0.0747	43.0200	--	--	--
R1	0.8107	0.0429	7.1580	-87.0%	-42.5%	-83.4%
R2	0.7453	0.0434	7.0842	-88.1%	-41.9%	-83.5%
R3	0.7517	0.0436	7.0879	-88.0%	-41.7%	-83.5%
R4	0.7615	0.0438	7.0907	-87.8%	-41.4%	-83.5%
R5	0.7715	0.0439	7.0931	-87.6%	-41.2%	-83.5%
R6	0.7817	0.0441	7.0955	-87.5%	-40.9%	-83.5%
R7	0.7920	0.0443	7.0979	-87.3%	-40.7%	-83.5%
R8	0.8024	0.0445	7.1002	-87.1%	-40.4%	-83.5%
R9	0.8128	0.0447	7.1024	-87.0%	-40.1%	-83.5%
R10	0.8101	0.0447	7.0445	-87.0%	-40.1%	-83.6%

The large IC-to-R1 drops (N: -87%, C: -84%) represent IC spin-down from unrealistic initial conditions. From R1 onward, the totals stabilize:

- Total N slowly increasing (NO₃ accumulation without denitrification)
- Total P slowly decreasing (POP settling loss)
- Total C slowly increasing (DIC equilibrating with boundary)

15. Process-Level Root-Cause Analysis Summary

Zooplankton N/P Collapse

Severity: CRITICAL

Symptom: ZOO_N and ZOO_P = 0 in all boxes after Year 1

Root Cause:

Three compounding causes:

1. Boundary conditions have ZOO_N = ZOO_P = 0 (FORC_TS files)
2. Variable stoichiometry mode (ZOOP_OPTION_1=1) has no homeostatic N:C floor
3. Excretion terms R_ZOO_EX_DON/DOP never computed (developer bug note in source)

Recommendation:

Set ZOO_N = ZOO_C * 0.22 and ZOO_P = ZOO_C * 0.024 in all FORC_TS files. Consider switching to ZOOP_OPTION_1=0 or adding min N:C enforcement.

Phytoplankton Year-over-Year Decline (~4%/yr)

Severity: WARNING

Symptom: DIA_C, CYN_C, OPA_C all show slow annual decline in end-of-year and peak values

Root Cause:

Combination of:

1. Boundary dilution (Baltic DIA_C = 0.01 vs internal ~1.3 peak)
2. Settling loss without sediment recycling (MODEL_SEDIMENTS=0)
3. Winter mortality floor (KD_DIA_20 = 0.12/day) acts as seasonal ratchet

Recommendation:

Enable sediment model (MODEL_SEDIMENTS=1) for benthic recycling. Consider reducing KD_DIA_20 or adding seed population mechanism.

NO3 Accumulation (+1.5%/yr)

Severity: WARNING

Symptom: Nitrate slowly increasing in all boxes without reaching steady state

Root Cause:

No denitrification pathway active:

1. MODEL_SEDIMENTS=0 disables sediment denitrification
2. No pelagic denitrification process in current configuration
3. Nitrification converts NH4 -> NO3 but no reverse pathway

Recommendation:

Enable sediment model for sediment denitrification. Or consider adding pelagic denitrification process.

IC Spin-Down (Year 1)

Severity: INFO

Symptom: All variables show large IC-to-R1 transition as unrealistic ICs wash out

Root Cause:

Initial conditions are rough estimates, not at equilibrium. This is expected and the purpose of multi-year spin-up runs.

Recommendation:

Use R10 final state as ICs for future simulations (write MTRX files). Consider using 5+ repeat spin-up before analysis runs.

16. Recommendations & Next Steps

Immediate: Fix Zooplankton Boundary Conditions

Update all FORC_TS files to set $ZOO_N = ZOO_C * 0.22$ and $ZOO_P = ZOO_C * 0.024$. This is the minimum fix to prevent N/P decoupling from carbon. Re-run simulation to verify zooplankton stoichiometry recovery.

Short-term: Enable Sediment Model

Set `MODEL_SEDIMENTS=1` in the input file to activate benthic recycling. This will provide: (a) denitrification pathway to stabilize NO_3 , (b) nutrient return from settled detritus to support phytoplankton, (c) sediment oxygen demand for realistic summer DO depression.

Short-term: Use R10 State as New Initial Conditions

The MTRX files contain the final state vector. These can be used as initial conditions for future runs to skip the spin-up phase. This is standard practice for ecological models.

Medium-term: Review Zooplankton Source Code

Fix the `R_ZOO_EX_DON / R_ZOO_EX_DOP` computation (currently never assigned). Add a minimum N:C ratio enforcement to prevent stoichiometric collapse. Consider whether `ZOOP_OPTION_1=0` (fixed stoichiometry) is more appropriate.

Long-term: Calibration Campaign

With the structural issues fixed, conduct a formal calibration using observed data to tune: `KG_DIA_OPT_TEMP`, `KD_DIA_20`, `KG_ZOO_OPT_TEMP`, settling velocities, and half-saturation constants. The 10-year spin-up run provides a stable baseline for parameter sensitivity analysis.

Appendix A: Simulation Configuration

Input file:	INPUT_3560day.txt
Simulation start:	6209.0 Julian days
Simulation end:	6565.0 Julian days
Number of repeats:	10
Effective duration:	3560 days (9.75 years)
Time steps per day:	240
Print interval:	240 (1 output per day)
Output boxes:	5, 6, 8, 9, 14, 17, 25
Sediment model:	OFF (MODEL_SEDIMENTS=0)
Resuspension:	OFF (RESUSPENSION_OPTION=0)
WCONST file:	WCONST_04.txt (with detritus C:N fix)

Key WCONST_04.txt Parameters

Diatoms:	KG=3.7/d, KD=0.12/d, KR=0.05/d, EFF=0.95, I_S=100, KHS_DIN=0.01, KHS_DIP=0.005, KHS_DSi=0.013
Cyanobacteria:	KG=1.0/d, KD=0.10/d, KR=0.02/d, EFF=0.90
Other Phyto:	KG=2.0/d, KD=0.10/d, KR=0.03/d, EFF=0.90
Zooplankton:	KG=0.45/d, KD=0.15/d, KR=0.03/d, EFF=0.80, FOOD_MIN=0.02, N:C=0.22, P:C=0.024
Detritus:	KDISS_C=0.25/d (fixed from 10.0), FAC_PHYT_C=0.0 (fixed from 2.0)